

MANGALORE REFINERY & PETROCHEMICALS LIMITED

PROCESS ENGINEERING, ROTATING MACHINERY & ASSET INTEGRITY DIRECTORATE

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COMPREHENSIVE ENGINEERING & OPERATIONS MASTER MANUAL

API 610 CENTRIFUGAL PUMPS & REFINERY PROCESS UNITS

COMPLEX 01-08: CDU/VDU, DHDT, FCCU, HGU, SRU, PPU, SPM & DESALINATION

DOCUMENT IDENTIFIER: MRPL-ENG-MAN-API610-2026-REV5.0
COMPLEX SCOPE: UNITS 01 TO 08 (ATMOSPHERIC, HYDROTREATER, CRACKER, HYDROGEN, SULFUR)
GOVERNING CODES: API 610 (12th Ed), API 682 (4th Ed), API 617, API 618, ASME VIII, B31.3
STATUTORY MANDATES: OISD-STD-118, OISD-STD-240, PNGRB REGULATIONS, FACTORY ACT 1948
SAFETY INTEGRITY: IEC 61508 / IEC 61511 SIL-2 / SIL-3 2003 TRIPLE MODULAR REDUNDANT SIS
GOVERNANCE STANDARD: MANDATORY FOUR-EYES DUAL SUPERVISORY SIGN-OFF (LEVEL 4 CHIEF ENGINEER)
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DOCUMENT GOVERNANCE & MULTI-DISCIPLINARY APPROVAL RECORD

DISCIPLINE / ROLE	AUTHORITY & EMP ID	STATUS	DATE
Rotating Equipment Lead:	Er. R. K. Shenoy (EMP-3841)	VERIFIED	2026-08-20
Chief Safety & Loss Prevention:	Er. Jane D'Souza (EMP-7102)	APPROVED	2026-08-25
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Statutory OISD Inspectorate:	Directorate of Boilers	VALIDATED	2026-09-02

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STATUTORY COMPLIANCE & REGULATORY FRAMEWORK

Mandatory compliance matrix across OISD, API, ASME, and IEC standards

REGULATORY STANDARDS HIERARCHY
ENFORCEMENT POLICY & LEGAL SANCTIONS
Section 3.1 of OISD-STD-118 specifies that deviation from certified operating limits without an approved Management of Change (MOC) constitutes a statutory violation under the Indian Petroleum Act 1934 and Factory Act 1948. All bypass valves, relief valve isolations, and trip interlock overrides require Tier-4 Chief General Manager counter-signature. Unsanctioned bypasses will initiate immediate automatic DCS safe shutdown.
CRITICAL STATUTORY NOTICE
All field procedures detailed herein are legally binding on shift engineers, board operators, and maintenance contractors. Any discrepancy between field conditions and this manual must be halted immediately and reported to CCR.

REFINERY COMPLEX ARCHITECTURE OVERVIEW

Integrated unit layout, throughput capacities, and interconnecting pipelines

MRPL REFINING UNITS MASTER SPECIFICATION
INTERCONNECTING HEADER NETWORK
The refinery operates continuous 24-inch and 36-inch crude transfer headers from NMPT SPM directly to the Crude Oil Tank Farm (COT). High-pressure hydrogen is circulated at 45.0 barg via the Unit 04-02-03 ring main. Sour water and acid gas networks are isolated under nitrogen purge with continuous H2S monitoring.
PROCESS INTERDEPENDENCY WARNING
Loss of Hydrogen header pressure from Unit 04 will trip DHDT Unit 02 within 45 seconds to prevent catalyst coking. Unit 02 booster pump P-101A will auto-trip on low reactor feed flow.

UNIT 01: CRUDE & VACUUM DISTILLATION (CDU/VDU)

Atmospheric fractionator, vacuum column, furnace duty, and cut points

CDU/VDU OPERATING PARAMETERS & BOUNDARY ENVELOPE

DISTILLATION CUT-POINT SPECIFICATION
- Light Naphtha (IBP - 90 deg C): Routed to ISOM unit for RON enhancement. - Heavy Naphtha (90 - 140 deg C): Feedstock to HGU (Unit 04) and Catalytic Reformer. - Superior Kerosene / ATF (140 - 240 deg C): Merox treated for Jet A-1 aviation turbine fuel. - Gas Oil (240 - 370 deg C): Pumped via booster P-101A directly to DHDT Unit 02 for desulfurization. - Vacuum Gas Oil (VGO 370 - 560 deg C): Cracked in PFCCU Unit 03 into high-octane gasoline. - Vacuum Residue (>560 deg C): Processed in Visbreaker / Bitumen Blowing Unit.

CRUDE CHARGE WATER SLUG HAZARD
Water slugs entering the atmospheric furnace will flash instantaneously (1,600x volume expansion). Maintain desalter brine interface level LT-1025 within 35-55% at all times.

UNIT 02: DIESEL HYDROTREATER (DHDT) COMPLEX

Ultra-low sulfur diesel production, reactor loop, and booster train

DHDT HIGH-PRESSURE REACTOR CIRCUIT SPECIFICATIONS

DESULFURIZATION CHEMISTRY & PRODUCT PURITY
The trickle-bed catalytic reactor utilizes CoMo/NiMo on alumina catalyst to reduce sulfur from 12,000 ppm to < 8.0 ppm (BS-VI Euro-VI specification). Reaction: Organic Sulfur + H2 -> Hydrocarbons + H2S. Wash water must be injected continuously upstream of the combined feed/effluent exchanger to prevent ammonium bisulfide (NH4HS) crystallization and catastrophic plugging.

HOT HYDROGEN ATTACK & EMBRITTLEMENT
Operating above Nelson Curve limits for 2.25Cr-1Mo metallurgy induces methane bubble void formation. Hydrogen partial pressure must never exceed 82.0 barg at temperatures above 380 deg C.

UNIT 03: FLUID CATALYTIC CRACKING UNIT (PFCCU)

Catalyst circulation, riser reactor, regenerator cyclone, and slide valves

PFCCU KEY OPERATING ENVELOPE

CATALYST CIRCULATION & SLIDE VALVE PROTECTION
PFCCU relies on Zeolite Y catalyst micro-spheres circulating between reactor and regenerator. Hydraulic slide valves control catalyst flow. In the event of differential pressure reversal (< 0.25 bar), the non-return check valves and emergency purge steam will actuate within 1.5 seconds to prevent explosive hydrocarbon ingress into the air regenerator.

AFTERBURNING EXPLOSION HAZARD
Sudden CO afterburning in regenerator cyclones will melt internal cyclone diplegs (>800 deg C). Immediately inject quench steam if cyclone outlet delta-T exceeds 25 deg C over the bed.

UNIT 04: HYDROGEN GENERATION UNIT (HGU)

Steam methane/naphtha reforming, shift conversion, and PSA purification

HGU PROCESS CONDITIONS & TECHNICAL SPECIFICATIONS

STEAM METHANE/NAPHTHA REFORMING PRINCIPLES
Reaction: $C_nH_m + n H_2O \rightleftharpoons n CO + (n + m/2) H_2$ (Endothermic at 880 deg C over Nickel catalyst). Water Gas Shift: $CO + H_2O \rightleftharpoons CO_2 + H_2$ (Exothermic at 350 deg C over Iron-Chrome catalyst). Pressure Swing Adsorption (PSA): 10-bed molecular sieve adsorption extracting 99.99% pure H2 while discharging tail gas (CO, CO2, CH4) back to reformer burners as fuel.

CATALYST CARBON DEPOSITION (COKING)
Operating with Steam-to-Carbon ratio below 2.80 causes rapid filament carbon growth on nickel reforming catalysts, leading to localized tube hot spots and explosive tube rupture within 4 hours.

UNIT 05: SULFUR RECOVERY UNIT (SRU & TGTU)

Claus reaction furnace, catalytic converter, and tail gas incineration

SRU ENVIRONMENTAL & COMBUSTION CONTROL

MODIFIED CLAUS CHEMISTRY & DESTRUCTION OF CONTAMINANTS

Thermal Stage: $\text{H}_2\text{S} + 1.5 \text{ O}_2 \rightarrow \text{SO}_2 + \text{H}_2\text{O}$ (Destroys NH_3 and BTX at $>1150 \text{ deg C}$).

Catalytic Stage: $2 \text{ H}_2\text{S} + \text{SO}_2 \leftrightarrow 3 \text{ S} + 2 \text{ H}_2\text{O}$ (Titanium/Alumina catalyst at 230 deg C).

Tail Gas Treatment Unit (TGTU) hydrogenates residual SO_2 back to H_2S for MDEA absorption, achieving 99.9% overall sulfur recovery efficiency conforming to CPCB standards.

TOXIC H2S BACK-PRESSURE & SEAL BLOWOUT

Liquid sulfur seal legs prevent toxic acid gas from entering the sulfur pit. Never allow seal leg steam heating to fail. Solidified sulfur causes instant pressure surges that rupture furnace rupture discs.

UNIT 06: POLYPROPYLENE POLYMER PLANT (PPU)

Propylene purification, loop reactor, fluid-bed gas phase, and extrusion

PPU HIGH-PRESSURE POLYMERIZATION PARAMETERS

ZIEGLER-NATTA POLYMERIZATION DISCIPLINE
High-activity 4th generation Ziegler-Natta titanium catalyst produces homopolymers and random copolymers. TEAL co-catalyst is pyrophoric (ignites spontaneously in air) and must be handled under pure nitrogen blanket (>99.999% N2). Emergency catalyst kill system injects CO / ATMER inhibitor within 0.8 seconds if loop reactor temperature exceeds 78 deg C.

RUNAWAY EXOTHERMIC POLYMERIZATION
Polymerization of propylene releases 2,400 kJ/kg. If circulation pump P-6001 trips, static resin will melt into a solid block of plastic within 3 minutes, permanently destroying the loop reactor.

UNIT 07: SINGLE POINT MOORING (SPM) & CRUDE TERMINAL

Offshore marine loading, subsea pipelines, PLEM, and tank farm

SPM MARINE OFFLOADING SPECIFICATIONS

PIPELINE END MANIFOLD (PLEM) & SUBSEA INTEGRITY
Crude is transferred via twin 48-inch subsea lines to the 1.2 million cubic meter Crude Oil Tank Farm. Intelligent pigging with Magnetic Flux Leakage (MFL) and Ultrasonic Testing (UT) pigs is conducted every 36 months to verify wall integrity against marine microbiologically induced corrosion (MIC).

OIL SPILL RESPONSE (TIER-1 / TIER-2)
Marine breakaway couplings (MBC) activate under excessive wave action (>3.5 m swell) or pressure surges (>18.5 barg) to prevent marine oil pollution. Continuous radar satellite surveillance is mandated by Indian Coast Guard.

UNIT 08: DESALINATION & EFFLUENT TREATMENT

Sea water reverse osmosis (SWRO), brine dispersion, and zero liquid discharge

30 MLD SEA WATER DESALINATION OPERATIONAL CRITERIA

ZERO LIQUID DISCHARGE & SUSTAINABLE REFINERY WATER BALANCE
The Tannirbhavi SWRO plant provides 30 million liters per day, decoupling MRPL from the Netravathi River. Permeate water undergoes decarbonation and mixed-bed polishing to supply boiler feed water (<0.1 uS/cm) for 110 bar superheated steam generation in captive power boilers.

MEMBRANE FOULING & CHLORINE OXIDATION
RO polyamide membranes are permanently destroyed by free chlorine. Sodium bisulfite (SBS) dosing must maintain Oxidation-Reduction Potential (ORP) strictly below +200 mV upstream of HP membranes.

ROTATING MACHINERY CLASSIFICATIONS (API 610)

Overhung, between-bearings, and vertically suspended pump architectures

API 610 12TH EDITION PUMP CONFIGURATION MATRIX

ENGINEERING SELECTION CRITERIA FOR HEAVY GASOIL
For temperatures exceeding 200 deg C and pressures exceeding 25 bar, API 610 mandates radially split centerline-mounted casings (BB2 or BB5) to ensure symmetric thermal growth and prevent piping nozzle stress misalignment. The P-101A/B train utilizes a BB5 double-casing barrel pump designed to withstand 500 PSI MAWP at 260 deg C.

CENTERLINE MOUNTING INTEGRITY
Never tighten pump casing hold-down bolts while the casing is hot without verifying sliding pin clearance. Binding at the centerline support keys results in shaft deflection, seal face distortion, and bearing seizure.

PUMP METALLURGY & HYDROTEST LIMITS

ASTM material classes, impeller trim schedules, and casing proof tests

API 610 MATERIAL CLASS DEFINITIONS FOR REFINERIES

P-101A MATERIAL CERTIFICATE & PROOF PRESSURE
- Pump Tag: P-101A / P-101B (Heavy Gasoil Hydrotreater Booster) - Material Class: API 610 Class S-6 (CA6NM 12% Chrome Impeller / WCB outer barrel with 316L weld overlay) - Maximum Allowable Working Pressure (MAWP): 500.0 PSI (34.5 barg) at 260 deg C - Factory Hydrostatic Shell Test: 750.0 PSI (1.5x MAWP) held for 60 minutes with zero leakage - Impeller Trim: Rated Diameter 342 mm (Maximum: 360 mm, Minimum: 315 mm)

CAVITATION EROSION ON 12% CR IMPELLERS
Operating below Minimum Continuous Stable Flow (MCSF 120 m3/h) induces suction recirculation that pits CA6NM blades. Inspect impeller inlet edges every turnaround via dye penetrant testing.

MECHANICAL SEAL STANDARDS (API 682 4TH ED)

Single and dual unpressurized arrangements: Plan 11, Plan 23, and Plan 52

API 682 SEAL PIPING PLANS OVERVIEW

PLAN 23 THERMODYNAMIC DUTY FOR HOT HYDROCARBONS
Plan 23 uses a high-efficiency bi-directional pumping ring mounted on the shaft to circulate seal chamber fluid through a dedicated shell-and-tube seal cooler. This isolates the dynamic silicon carbide faces from bulk fluid temperatures (220 deg C down to 65 deg C in seal cavity), preventing hydrocarbon vaporization across the seal interface.

AIR ENTRAINMENT IN PLAN 23 COOLERS
High-point vent valves on Plan 23 seal coolers must be thoroughly vented during initial fluid priming. Air pockets insulate the cooling tubes, causing seal face temperatures to exceed 180 deg C within 60 seconds.

PLAN 53A / 53B BARRIER SYSTEMS MAINTENANCE

Barrier fluid selection, nitrogen pressurization, and bladder accumulator checks

API 682 PLAN 53A BARRIER SYSTEM OPERATING RULES			
PARAMETER / COMPONENT	TARGET OPERATING VALUE	ALARM LOW (PAL)	EMERGENCY TRIP
-			
Barrier Fluid Type	Synthetic Polyalphaolefin (ISO VG 32 Synthetic Turbine Lube)		
Barrier Reservoir Pressure	45.0 - 50.0 PSI ABOVE Max Seal Chamber Pressure (PT-5301)		
Minimum Overpressure Margin	Strictly 1.5 bar (22 PSI) over inner seal cavity pressure		
Barrier Pot Liquid Level (LT-53)	55.0 - 75.0 %	35.0 % (Alarm Low)	20.0 % (Trip)
P-101A)			
Barrier Loop Temperature (TT-53)	35.0 - 55.0 deg C	65.0 deg C (Cooler Clean)	80.0 deg C
Nitrogen Supply Regulator (PCV)	12.5 barg Regulated	10.0 barg (N2 Loss)	8.0 barg (Safe Interlock)

BARRIER FLUID CONSUMPTION & LEAK DETECTION LOGIC
In Plan 53A, normal seal face lubrication consumes 5 to 15 mL/day of barrier fluid across the inner face into the process.
- Rapid level drop (>50 mL/h): Indicates inner seal face blowout. Barrier fluid is leaking into process.
- Rapid level rise: Indicates catastrophic inner face failure where high-pressure process fluid backflows into the barrier pot, contaminating the reservoir. Immediately trip pump!

REVERSE PRESSURE BLOWOUT HAZARD
If barrier pressure drops below seal chamber pressure, the inner seal faces will be pushed apart by process fluid, causing immediate toxic hydrocarbon leakage to the atmospheric secondary seal.

API 617 / 618 HEAVY GAS COMPRESSORS

Centrifugal recycle compressors, reciprocating make-up machines, and anti-surge

COMPRESSOR MECHANICAL & AERO-THERMODYNAMIC LIMITS

ANTI-SURGE CONTROL LOOP DYNAMICS
Centrifugal compressor surge occurs when inlet flow drops below the aerodynamic stability boundary, causing high-frequency pressure oscillations (10-30 Hz) and violent axial thrust reversals. The digital anti-surge controller maintains a minimum 10% safety margin from the Surge Limit Line (SLL). If rate-of-approach exceeds threshold, the fast-acting anti-surge valve pops open in 750 milliseconds.

LIQUID CARRYOVER INTO SUCTION SCRUBBER
Liquid droplets entering high-speed centrifugal impellers cause immediate blade erosion and devastating rotor unbalance. Suction knockout drum demister level LT-2010 has SIL-3 2oo3 interlock tripping K-201.

PROCESS INSTRUMENTATION & SIGNAL ARCHITECTURE

Pressure, temperature, flow, and level transmitters across Honeywell/Yokogawa DCS

MASTER FIELD INSTRUMENTATION SIGNAL MATRIX

REDUNDANCY & TRANSMITTER HEALTH MONITORING
Safety-critical transmitters utilize NAMUR NE 43 diagnostic signal thresholds: - Signal < 3.6 mA: Severe Sensor Burnout / Open Circuit Fault (Fail-Safe Low) - Signal 3.8 to 20.5 mA: Valid Measurement Range - Signal > 21.0 mA: Sensor Short-Circuit / Over-Range Fault (Fail-Safe High) Transmitters failing outside NAMUR bands generate instant high-priority instrument trouble alarms on DCS.

IMPULSE LINE FREEZING & BLOCKAGE
Heavy gasoil impulse lines require continuous steam tracing and winterization. Sluggish transmitter response (>1.5s delay) indicates wax deposition. Blow down impulse lines using clean kerosene solvent.

SAFE OPERATING ENVELOPES (SOE) & BOUNDARY CONTROLS

Multi-variable process boundaries, operating margins, and integrity limits

MULTI-VARIABLE SAFE OPERATING BOUNDARIES			
BOUNDARY REGION	DESCRIPTION	CONTROL ACTION	GOVERNANCE

Normal Envelope Operator	Standard Operating Zone (SOZ)	Automated DCS PID Controllers	Shift
Alert Region Operator Log	Operating Window Excursion (OWE)	Alarms Sound, Operator Adjusts	Board
Critical Envelope In-Charge Required	Integrity Operating Limit (IOL)	Automatic Safeguard Pre-Trips	Shift
Unsafe Zone Investigation	Safety Instrumented Trip (SIS)	Emergency Shutdown System Triggers	Incident

SOE COORDINATE TABLE FOR UNIT 01 BOOSTER PUMP P-101A

SIMULTANEOUS MULTI-VARIABLE EXCURSION

If both discharge pressure exceeds 440 PSI AND vibration exceeds 4.0 mm/s, the operating point has entered the hydrodynamic stall envelope. Do not throttle discharge valve; immediately cycle standby pump.

ISA-18.2 ALARM RATIONALIZATION & MANAGEMENT

Alarm priorities, suppression of chattering tags, and shelving protocols

ISA-18.2 ALARM PRIORITY CLASSIFICATION STRUCTURE			
PRIORITY TIER	COLOR CODE	RESPONSE TIME REQUIREMENT	OPERATOR ACTION REQUIRED

CRITICAL (P1) Initiation	Flashing Red	Immediate (< 1 Minute)	Emergency Trip Intervention / Safe Stop
HIGH (P2) Lineup	Amber Yellow	Rapid (< 5 Minutes)	Corrective Setpoint Adjustment / Valve
MEDIUM (P3) Monitoring	Light Blue	Standard (< 15 Minutes)	Field Inspection / Routine Process
LOW (P4) Notification	White / Grey	Informational (< 1 Hour)	Shift Handover Log Entry / Maintenance

ALARM SHELIVING & CHATTERING SUPPRESSION ALGORITHMS
- Chattering Alarms: Any alarm triggering more than 3 times in 60 seconds is automatically moved to temporary deadband suppression (+/- 2.5% span) to prevent operator cognitive fatigue. - State-Based Alarm Shelving: When Unit 01 is in 'Standby' or 'N2 Purge' mode, all process low-flow and low-pressure alarms are automatically suppressed by the DCS state engine. - Maximum Shelving Duration: Strictly 8 hours. Requires written Four-Eyes supervisor authorization.

ALARM FLOOD THRESHOLD EXCEEDED
EEMUA 191 / ISA-18.2 benchmarks dictate maximum 10 alarms per 10 minutes per console. An alarm flood (>30 alarms/10 min) requires CCR Lead Engineer to invoke Emergency State Protocol.

EMERGENCY SHUTDOWN (SIS) & 2003 VOTING LOGIC

IEC 61511 Safety Instrumented Functions, interlocks, and voting logic

SAFETY INSTRUMENTED INTERLOCK SCHEDULE (P-101A/B)

2003 (TWO-OUT-OF-THREE) VOTING ARCHITECTURE
In 2oo3 voting, three independent transmitter channels (A, B, and C) measure the identical physical parameter. - If 1 transmitter detects an excursion: An alarm sounds, and an 'Instrument Discrepancy' alert is flagged. - If 2 or 3 transmitters detect an excursion: The Safety PLC executes the emergency trip instantaneously. This architecture eliminates nuisance false trips while maintaining 99.999% hardware fault tolerance.

MANUAL SIS BYPASS STRICTLY PROHIBITED
Bypassing any SIL-3 2oo3 interlock key without an authorized physical key-switch lock and Tier-4 Management of Change (MOC) will trigger a sovereign security audit violation and automatic CCR trip.

OVERPRESSURE RELIEF VALVES & FLARE SYSTEM

ASME Section VIII safety relief valves (SV-101 to SV-405) and flare headers

SAFETY RELIEF VALVE (PSV) DESIGN DATA SHEET

RELIEF VALVE MAINTENANCE & IN-SITU POP TESTING
- Testing Frequency: All refinery process relief valves must undergo physical bench testing every 24 months. - Car-Seal Protocol: Isolation block valves upstream and downstream of PSVs are strictly Car-Sealed Open (CSO). - Rupture Disc Assembly: Installed upstream of toxic sour gas relief valves to eliminate fugitive emissions. - Flare KO Drum: Equipped with dual duty/standby auto-start pumps to prevent liquid carryover to flare tip.

CLOSED BLOCK VALVE ON RELIEF LINE
Operating any process vessel with its relief block valve closed or unsealed constitutes a Tier-1 process safety hazard. Instant dismissal and statutory regulatory prosecution apply under OISD regulations.

EMERGENCY DEPRESSURIZATION (BLOWDOWN) SYSTEM

Rapid depressurization valve matrix (XV-101 to XV-108) and flare routing

EMERGENCY DEPRESSURIZATION (EDP) SCHEDULE

DEPRESSURIZATION KINETICS & COLD METAL TEMPERATURE (J-T EFFECT)
Rapid depressurization of high-pressure hydrogen and hydrocarbon vapor causes severe Joule-Thomson cooling. Process lines can plummet from +350 deg C down to -40 deg C within 15 minutes. All emergency blowdown lines are fabricated from ASTM A333 Grade 6 low-temperature impact-tested carbon steel to prevent brittle catastrophic fracture during full depressurization.

FLARE BACKPRESSURE SURGE DURING BLOWDOWN
Simultaneous depressurization of multiple units will overwhelm the flare header. Staggered blowdown logic is hardcoded into the Triconex safety system with prioritized interlocking.

PRE-COMMISSIONING & COLD LOOP CHECKS

Mechanical completion sign-off, loop checks, and electrical cold checks

PRE-COMMISSIONING MULTI-DISCIPLINARY CHECKLIST
ELECTRICAL SAFETY & LOCKOUT VERIFICATION
Motor breaker racking at the 6.6 kV substation must follow strict NFPA 70E Arc Flash and OISD-STD-240 protocols. Field stop pushbuttons must be tested in the uncoupled state. Verify that emergency trip response at the local field control station registers in CCR in less than 250 milliseconds.
UNVERIFIED COLD ROTATION
Never couple motor to pump before confirming motor uncoupled phase rotation direction. Running a centrifugal pump backwards will unscrew the impeller from the shaft, destroying the casing within 5 seconds.

NITROGEN PURGING & INERTIZATION PROTOCOL

System oxygen evacuation, sweep purging, and explosive mixture prevention

NITROGEN PURGE & OXYGEN EVACUATION PROCEDURE				
PHASE	ACTION / METHOD	PRESSURE TARGET	OXYGEN TARGET	COMPLETION CRITERIA

Phase 1	Atmospheric Venting cessation	0.1 barg	Air Displaced	Visual steam/air
Phase 2	Pressure Sweep Purge (N2) pressurize/depress	6.0 barg	< 1.0 vol % O2	Repeat 4 cycles of
Phase 3	Continuous Vacuum Sweep correlation	-0.8 barg	< 0.5 vol % O2	Helium leak test
Phase 4	Final Blanket Pressurization	1.5 barg (Pure N2)	< 0.2 vol % O2	Hold pressure for 4 hours with 0 delta-P

ATMOSPHERIC OXYGEN TESTING & GAS CHROMATOGRAPHY
Portable calibrated optical oxygen analyzers (sampling points SP-101 through SP-108) must confirm that residual oxygen in all vessels, casing cavities, and suction headers is strictly below 0.2 vol % (2,000 ppm) prior to introducing any hydrocarbons or hydrogen. Gas chromatography verification is mandatory.

CONFINED SPACE NITROGEN ASPHYXIATION
Nitrogen produces instant unconsciousness and asphyxiation with zero warning odor. Warning signs and barricades are mandatory around all open purge vents. Never enter a vessel without positive fresh-air breathing gear.

CHEMICAL CLEANING, PASSIVATION & LUBE FLUSHING

Pickling, neutralizing, citric acid passivation, and lube oil filtration

PIPING CHEMICAL CLEANING & PASSIVATION STAGES				
STAGE	CHEMICAL SOLUTION	TEMPERATURE	CIRCULATION TIME	PURPOSE / OBJECTIVE

Stage 1	Alkaline Degreasing (1.5% NaOH)	75 - 85 deg C	12 - 16 Hours	Remove oils, greases, preservatives
Stage 2	Demineralized Water Rinse	Ambient	Until pH < 8.5	Flush alkaline residues
Stage 3	Acid Pickling (3.0% Citric Acid)	60 - 70 deg C	8 - 12 Hours	Dissolve mill scale & rust
Stage 4	Iron Neutralization & Passivation	50 - 60 deg C	6 - 8 Hours	Establish magnetite protective layer
Stage 5	Final Hot Dry Nitrogen Dehydration	90 - 110 deg C	Until Dewpoint <-40 deg C	Eliminate moisture pockets

LUBE OIL SYSTEM FLUSHING & MICRO-FILTRATION
High-velocity hot oil flushing is conducted using external 5-micron filtration carts. Flushing velocity must achieve a Reynolds number Re > 4,000 to ensure turbulent dislodgement of weld slag and grit. Flushing continues until 100-mesh stainless steel inspection screens show zero metallic particles over 4 hours.

CHLORIDE CONTAMINATION IN STAINLESS CIRCUITS
Total chloride content in cleaning solutions, rinse water, and passivating chemicals must not exceed 25 ppm. High chlorides initiate stress corrosion cracking (SCC) in austenitic stainless steels.

PUMP PRIMING & SUCTION DE-AERATION

Casing liquid filling, high-point venting, and minimum flow bypass warmup

PRIMING & SUCTION VALVE LINEUP DISCIPLINE
VAPOR LOCK ELIMINATION & TEMPERATURE EQUALIZATION
Cold pumps handling hot gasoil (>180 deg C) must undergo controlled warm-up via a cracked warm-up bypass valve. Heating rate must not exceed 1.5 deg C per minute to avoid casing thermal distortion and mechanical seal face warping. Never start a pump if the casing top-to-bottom temperature differential exceeds 20 deg C.
DRY RUNNING MECHANICAL SEAL SEIZURE
Starting a pump with vapor trapped in the seal chamber causes dry contact between silicon carbide faces. Faces will shatter from thermal shock within 12 seconds of motor start.

COLD STARTUP SEQUENCE (P-101A/B)

Step-by-step procedure for high-pressure heavy gasoil booster pumps

STEP-BY-STEP COLD STARTUP PROTOCOL

MOTOR STARTING FREQUENCY LIMITATIONS
High-voltage (6.6 kV) cage induction motors are limited to: - Maximum TWO consecutive starts from cold ambient conditions. - Maximum ONE start from hot running operating temperature. If a second start fails, the motor stator must cool for a mandatory 45 minutes before attempting any re-energization.

DEAD-HEADING RUN WITH CLOSED DISCHARGE
Operating P-101A against a completely closed discharge valve without minimum flow bypass will boil the internal fluid within 45 seconds, resulting in explosive vapor pressure escalation and casing rupture.

HOT RESTART & THERMAL SHOCK PREVENTION

Re-starting standby equipment under hot refinery conditions

HOT RESTART OPERATIONAL RESTRICTIONS

THERMAL GROWTH & CLEARANCE CONSIDERATIONS
During hot restarts, the pump impeller and casing expand at different thermal rates. Improperly warmed pumps experience internal rubbing between the wear rings (CA6NM) and casing bushes. Minimum radial clearance must adhere to API 610 Table 6 (+0.125 mm over base design for high-temp service).

COLD FLUID SHOCK ON HOT CASING
Never introduce cold feed (<80 deg C) into a pump casing that is above 180 deg C. The immediate thermal gradient causes cast steel casings to crack through the volute tongue.

CONTINUOUS SHIFT SURVEILLANCE & ROUNDS

Operator hourly rounds, physical checks, stethoscope inspection, and log sheets

HOURLY OPERATOR SURVEILLANCE ROUTINE (LOG SHEET 01-A)

PHYSICAL WALKDOWN & HOUSEKEEPING STANDARDS

Operators must physically inspect the pump baseplate for seal leakage drips, loose foundation anchor bolts, oil mist lubricator pressure (maintain 20 inches H2O), and cooling water jacket flow rates.
Any hydrocarbon leak exceeding 3 drops per minute requires immediate maintenance tagging.

REPORTING ANOMALIES TO CENTRAL CONTROL ROOM

Field operators must immediately report any sudden jump in vibration (>1.0 mm/s increase in 10 minutes) or temperature (>5 deg C rise in 15 minutes) to CCR board operator for instant corroboration.

1+1 REDUNDANT PUMP AUTO-TRANSFER PROTOCOL

Duty/standby pump swap procedure, auto-start logic, and bi-weekly cycling

AUTO-TRANSFER INTERLOCK LOGIC & PERMISSIVES		
CONDITION / TRIGGER	AUTOMATED DCS RESPONSE	CYCLE TIMING

Duty Pump P-101A Discharge < 320 PSI	DCS initiates Standby P-101B Start	Auto-starts in < 2.5 seconds
Duty Pump P-101A Motor Trip	DCS initiates Standby P-101B Start	Auto-starts in < 1.8 seconds
Duty Pump P-101A Vibration > 7.0 mm/s	Auto-starts P-101B then trips P-101A	Bumpless process flow transfer
Manual Scheduled Maintenance Swap rotation	Operator ramps P-101B, dims P-101A	Bi-weekly scheduled rotation

BUMPLESS BI-WEEKLY PUMP ROTATION PROCEDURE
1.0 Confirm standby pump P-101B is fully primed, warmed up, and lined up with discharge valve cracked 5%. 2.0 Start P-101B motor from DCS console. Verify discharge pressure builds to 400 PSI. 3.0 Slowly open P-101B discharge MOV-102 while simultaneously throttling back P-101A discharge MOV-101. 4.0 Maintain total combined feed flow FT-104 constant within +/- 2.0 MT/h throughout the transfer. 5.0 Once P-101B reaches 100% flow, close P-101A discharge MOV-101 completely and de-energize motor.

SIMULTANEOUS TWO-PUMP OVERLOAD
Running both P-101A and P-101B at full flow simultaneously will overload the common downstream piping, exceeding the 500 PSI MAWP and lifting relief valve SV-101.

CONTROLLED PLANT SHUTDOWN & DEPRESSURIZATION

Normal unit shutdown, gradual cooling, slop rerouting, and nitrogen purging

CONTROLLED SHUTDOWN SEQUENCE (UNIT 01 & 02)

THERMAL CONTRACTION & FLANGE LEAK PREVENTION
During unit cooldown, differing thermal contraction coefficients between bolts and flanges can loosen joints. Piping maintenance teams must perform active hot-bolt retorquing at 150 deg C to prevent flammable leaks during the cooling trajectory. Continuous optical gas imaging (FLIR) is deployed across all unit manifolds.

VACUUM FORMATION DURING STEAM PURGE CONDENSATION
Closing block valves on vessels containing condensing steam will create an internal hard vacuum, collapsing thin-walled fractionator towers and surge drums like an aluminum can. Keep vacuum breakers open.

LONG-TERM EQUIPMENT PRESERVATION & N2 BLANKETING

Preservation of idle pumps, nitrogen blankets, and periodic shaft rotation

EQUIPMENT PRESERVATION PROTOCOL FOR INACTIVE UNITS		
COMPONENT / SUBSYSTEM	PRESERVATION MEDIUM	MAINTENANCE INSPECTION FREQUENCY

Pump Casing & Volute	Dry Nitrogen Gas (0.35 - 0.50 barg)	Daily pressure gauge check (PT-N2)
Bearings & Shaft Journal	Vapor Phase Corrosion Inhibitor	Weekly shaft manual rotation (1.25 turns)
Mechanical Seal Cavity	Barrier Fluid (Synthetic PAO)	Monthly level and cleanliness sample
Suction / Discharge Lines	Blinded with Spectacle Blinds (8-Fig)	Physical visual tag inspection
Electric Motor Stator	Space Heaters Continuously ON	Weekly insulation megger check (>50 MOhm)
Coupling Elastomer / Disc	Rust Preventative Coating Applied	Quarterly elastomeric inspection

SHAFT ROTATION DISCIPLINE (ANTI-BRINELLING)
Pumps idle for more than 7 days must have their shaft rotated by hand 1-1/4 turns (450 degrees) every Tuesday. This redistributes the weight of the rotor across the rolling element bearings, prevents false brinelling of bearing races, and redistributes protective lube oil across the journal surfaces.

NITROGEN BLANKET PRESSURE LOSS
Loss of nitrogen blanket pressure allows moist maritime air from the Arabian Sea to enter the equipment, causing severe atmospheric pitting and stress corrosion cracking within 14 days.

EMERGENCY: CAVITATION & SUCTION PRESSURE LOSS

Recognizing cavitation signatures, acoustic noise, and emergency countermeasures

CAVITATION IDENTIFICATION & DIAGNOSTIC SIGNATURES

IMMEDIATE COUNTERMEASURES FOR FIELD & CCR OPERATORS

1.0 Check feed surge drum level LT-108. If level is below 35%, immediately increase feed makeup from COT.

2.0 Verify suction strainer differential pressure PDT-1002. If high, cycle standby suction filter.

3.0 Throttle back discharge valve MOV-102 by 15% to move pump operating point back toward Best Efficiency Point (BEP).

4.0 If cavitation noise does not abate within 90 seconds, START STANDBY PUMP P-101B AND TRIP P-101A.

IMPELLER DESTRUCTION FROM CAVITATION

Cavitation bubble collapse micro-jets generate localized pressures up to 10,000 bar. Operating in severe cavitation for more than 15 minutes will erode the impeller shroud, causing catastrophic rotor seizure.

EMERGENCY: SEAL BARRIER FLUID FAILURE

Loss of Plan 53A barrier pressure, inner face rupture, and emergency trips

PLAN 53A FAILURE MODES & AUTOMATED RESPONSES		
ALARM / INDICATION	FAILURE MECHANISM	AUTOMATED INTERLOCK / ACTION

Pressure Switch PSL-5301 Trip LOW'	Nitrogen bottle depletion or leak	DCS Alarm 'SEAL BARRIER PRESS
Rapid Level Rise in Pot LT-53	Inner seal shattered (HC backflow)	Emergency Trip P-101A + Auto Standby
Rapid Level Drop in Pot LT-53	Outer seal shattered (Fluid leak)	Maintenance Alert; prepare pump swap
High Temperature in Loop TT-53	Pumping ring failure / cooler clog	Switch cooling water source to backup

EMERGENCY INTERVENTION SEQUENCE
If barrier fluid pressure drops below seal chamber pressure, high-pressure heavy gasoil will immediately force open the inner seal faces and blow into the barrier pot. - Step 1: Immediately hit EMERGENCY STOP on pump P-101A. - Step 2: Close barrier pot nitrogen supply valve to contain contaminated fluid. - Step 3: Evacuate the immediate pump bay area; deploy gas detectors to monitor volatile organic vapors. - Step 4: Line up fire water fog monitors to cover the pump seal gland in case of auto-ignition.

AUTO-IGNITION OF HOT GASOIL
Heavy gasoil at 220 deg C has an auto-ignition temperature of ~250 deg C in atmospheric air. Any pressurized spray contacting an un-insulated pipe flange will ignite instantly into a high-pressure jet fire.

EMERGENCY: REACTOR RUNAWAY EXOTHERM MITIGATION

Temperature runaway in hydrotreater bed, hydrogen quench, and emergency dump

EXOTHERM ALARM TIERS & AUTOMATIC RESPONSES

EXOTHERM PHYSICS & QUENCH GAS DYNAMICS

Hydrotreating reactions are strongly exothermic (-45 to -65 kJ/mol H2 consumed).
In the event of flow maldistribution or feed olefins spike, a localized thermal runaway can accelerate exponentially (>10 deg C/minute).
Hydrogen quench gas (cold H2 at 45 deg C) must be injected into the inter-bed redistributor to arrest the reaction wavefront before metallurgical yield occurs at 430 deg C.

METALLURGICAL CREEP RUPTURE AT HIGH TEMPERATURE

Operating 2.25Cr-1Mo reactor steel above 430 deg C under 85 barg pressure causes catastrophic hoop stress wall thinning and explosive rupture within minutes. Never hesitate to invoke full Emergency Depressurization.

EMERGENCY: TOXIC H2S GAS RELEASE CONTINGENCY

Hydrogen sulfide hazard, sensor mapping, wind sock directions, and ERP Tier-2

H2S TOXICITY LEVELS & EMERGENCY ACTION TIERS

REFINERY EVACUATION & INCIDENT COMMAND PROTOCOL
- Acoustic Sirens: Intermittent wavering siren indicates toxic gas release. - Evacuation Direction: Observe illuminated wind socks on Unit 01 and Unit 05 towers. Evacuate UPWIND and CROSSWIND to designated primary assembly muster point (Muster Area Alpha - Admin Gate 1). - Rescue Teams: Only certified Emergency Response Team (ERT) personnel wearing positive-pressure SCBA may enter the hot zone for personnel recovery and isolation valve shutoff.

NEVER ATTEMPT RESCUE WITHOUT SCBA
More than 50% of H2S fatalities are would-be rescuers entering toxic gas vapor clouds without breathing apparatus. Always sound the general alarm and wait for ERT containment.

FOUR-EYES DUAL-SUPERVISOR GOVERNANCE (OISD-240)

Mandatory dual-key authorizations, cryptographically signed audit logs

CRITICAL PLANT ACTIONS REQUIRING FOUR-EYES SIGN-OFF		
ACTION CATEGORY	ACTION DESCRIPTION	AUTHORIZATION LEVEL

CAT-A: SIS Bypass Manager + Safety Head	Overriding any SIL-2 or SIL-3 Safety Interlock	Level 4 Chief General
CAT-B: Vessel Entry + Gas Inspector	Permit-to-Work for Confined Space Entry	Shift Operations Manager
CAT-C: Flare Header Safety Head	Isolating any Relief Valve upstream block valve	DGM Operations + Plant
CAT-D: Boundary Swap Rotating Equipment Head	Overriding safe operating envelope trip threshold	Shift In-Charge +
CAT-E: High Pressure Area Supervisor	P-101A restart following overpressure excursion	Lead Rotating Engineer +

CRYPTOGRAPHIC MULTI-SIGNATURE WORKFLOW IN COGNISHIFT
All high-risk actions initiated via the CogniShift console generate a dual-signature approval token. - Primary Reviewer: Shift Field Engineer validates mechanical readiness and physical field checklist. - Secondary Reviewer: Area Chief Operations Supervisor verifies process safety conditions and logs private cryptographic key signature into SQLite audit store. Un-reviewed actions expire automatically after 15 minutes and fail closed.

SINGLE-USER AUTHORIZATION VIOLATION
Any attempt to bypass the Four-Eyes principle or falsify counter-signatures violates OISD-STD-240 Section 4.2. CogniShift automatically locks the operator console and alerts Plant Cyber-Security Cell.

DIGITAL LOCKOUT-TAGOUT (LOTO) ENERGY ISOLATION

Electrical, mechanical, hydraulic, and pneumatic isolation matrices

LOTO ZERO ENERGY STATE VERIFICATION MATRIX			
ENERGY TYPE	PRIMARY ISOLATION DEVICE	ISOLATION METHOD	VERIFICATION PROCEDURE

Electrical Start Button	6.6 kV Substation Breaker	Rack out breaker to 'TEST'	Visual disconnect + Try
Hydraulic	Suction & Discharge Valves	Close MOV-101/102 & Chain Padlock	Crack bleed valve; confirm 0.0 bar
Pneumatic	Instrument Air Supply	Close isolation valve & vent	Check actuator pressure gauge = 0
Thermal	Piping Heat Tracing	De-energize electric tracing	Infrared thermography scan < 40 deg C
Mechanical	Pump Shaft Impeller	Lockout shaft anti-rotation	Verify zero free-wheeling rotation

PHYSICAL LOCK BOX & PERMIT INTEGRATION
Individual technician padlocks are placed on a designated red Gang Lock Box. The key to the field isolation valves remains locked inside until the very last technician removes their personal tag and padlock. Testing for Zero Energy State must include physical push of the local field start button with electrician present.

RESIDUAL HYDRAULIC PRESSURE TRAP
Closing block valves on hot fluid lines traps hydraulic fluid between closed seats. As heat expands the fluid, thermal pressure can reach 2,000 PSI, blowing out bonnet gaskets when maintenance unbolts flanges.

CRYPTOGRAPHIC PERMIT-TO-WORK (PTW) ISSUANCE

Hot work, cold work, height work, excavation, and radiation permits

PERMIT-TO-WORK CLASSIFICATION & EXPIRATION CRITERIA				
PERMIT TYPE	COLOR CODE	MANDATORY CLEARANCE TESTS	VALIDITY DURATION	SIGN-OFF LEVEL

Hot Work Officer + Shift In-Chg	Red Border	100% LEL Check (<0.0%), Fire Hose	Shift Basis (8 Hrs)	Area Fire
Cold Work Maintenance Engineer	Green Border	Depressurized & LOTO Certified	7 Days Maximum	Area
Confined Space + Rescue Team Lead	Yellow Bord	O2: 19.5-23.5%, H2S: 0, CO: 0	Single Shift Only	Safety Officer
Height Work Maintenance Supervisor	Blue Border	Safety Harness & Scaffolding Tag	Shift Basis (8 Hrs)	Civil
Radiography Safety Officer	Purple Bord	Radiation Area Barricading 20 m	Night Shift (4 Hrs)	Radiological

DIGITAL GAS TESTING & CONTINUOUS MONITORING
Hot work permits require continuous explosive gas testing using infrared LEL monitors. If flammable gas exceeds 1.0% LEL, hot work must cease instantly. All permits are linked via QR code to the CogniShift digital registry. Physical copies must be laminated and posted on the job site board at all times.

EXPIRATION OF GAS CLEARANCE CERTIFICATE
Gas test certificates expire after exactly 2 hours of inactivity. If work is halted for lunch or shift handover, re-testing and counter-signature by certified gas tester are legally required before restarting.

HAZOP STUDY NODE 1: FEED SURGE & SUCTION HEADER

Hazard and Operability guide words, causes, consequences, and safeguards

NODE 1 HAZOP DEVIATION MATRIX (FEED SURGE DRUM V-101)

RECOMMENDATIONS & ACTION ITEMS
- Action Item 1.1: Install secondary independent guided-wave radar level transmitter (LT-108B) to establish SIL-3 2oo3 voting logic on drum low-level trip. - Action Item 1.2: Upgrade suction strainer differential transmitter PDT-1002 to smart HART protocol with automated flush sequence.

HAZOP SAFEGUARD INTEGRITY CHECK
Never operate Node 1 if more than one safeguard layer is in degraded status. Degradation of both level alarm and pump trip requires immediate unit throughput reduction by 50%.

HAZOP STUDY NODE 2: BOOSTER PUMP CASING & DISCHARGE

Discharge overpressure, check valve failure, vibration, and seal blowout

NODE 2 HAZOP DEVIATION MATRIX (BOOSTER PUMP P-101A)

NODE 2 RISK PRIORITIZATION & LOPA CLASSIFICATION
Node 2 is categorized as High Risk (Risk Priority Number RPN = 180) due to high operating pressure (385 PSI) and flammable hydrocarbon inventory. Independent Protection Layers (IPLs) include: (1) DCS Alarm PAH-101, (2) SIS Interlock I-101, and (3) ASME Section VIII Relief Valve SV-101.

CASE STUDY CITATION: 492.5 PSI EXCURSION
On August 14, 2026, transient throttling of MOV-102 caused PT-101 to spike to 492.5 PSI. SIS Interlock I-101 tripped the motor in 1.2 seconds, preventing casing burst. Maintain setpoints strictly.

HAZOP STUDY NODE 3: REACTOR PRE-HEAT EXCHANGERS

Shell-and-tube exchanger train, tube rupture, and high-pressure breakthrough

NODE 3 HAZOP DEVIATION MATRIX (EXCHANGER TRAIN E-101A/B)

TWO-THIRDS RULE (2/3 RULE) RELIEF VALVE SIZING
ASME Section VIII mandates that if the design pressure of the low-pressure side is less than 2/3 of the high-pressure side design pressure, a full tube rupture relief valve must be installed. Shell side SV-105 is sized for 100% supersonic gas breakthrough capacity in accordance with API 520 / 521.

HIGH PRESSURE TO LOW PRESSURE CONTAMINATION
A tube rupture event introduces 80 bar gasoil into a 15 bar system. The low pressure piping will explode within 800 milliseconds if isolation valve XV-107 fails to slam shut.

FAILURE MODE, EFFECTS & CRITICALITY ANALYSIS (FMECA)

Risk Priority Numbers (RPN), severity, occurrence, and detection ratings

FMECA QUANTITATIVE ANALYSIS FOR P-101A TRAIN

RPN MITIGATION STRATEGY & MAINTENANCE TASK MAPPING
- RPN > 80: Mandates daily operational rounds and continuous online sensor monitoring (DCS vibration & temp). - RPN 50-80: Covered by monthly vibration analysis and quarterly oil sampling. - RPN < 50: Addressed during scheduled 3-year turnaround overhaul cycles. Mechanical seal faces (RPN 96) have the highest operational risk and require strict Four-Eyes pre-start verification.

REVISED RPN TARGET POST-MODERNIZATION
Following installation of the CogniShift sovereign monitoring agent and automated interlocks, the calculated RPN for mechanical seal face failure was reduced from 144 down to 96.

ROOT CAUSE ANALYSIS (RCA) FAULT TREE: 492.5 PSI

Fault tree analysis and 5-Why investigation of the August 14 excursion

5-WHY INCIDENT INVESTIGATION TREE (REF: INC-2026-0814)
CORRECTIVE & PREVENTIVE ACTIONS (CAPA LOG)
- CAPA-01: Installed dual stainless steel weather hoods on all outdoor actuator linkages (Completed Aug 22). - CAPA-02: Modified DCS interlock logic to require manual confirmation before stroking MOV-102 during live runs. - CAPA-03: Integrated CogniShift autonomous telemetry auditor to flag valve stiction trends prior to excursions.
STATUTORY CLOSURE VERIFICATION
The Indian Boiler & Factory Inspectorate officially reviewed this RCA report on September 02, 2026, and cleared Unit 01 for unrestricted continuous operation at 410 PSI baseline.

LASER SHAFT ALIGNMENT STANDARDS (API 610 TAB 15)

Parallel and angular alignment tolerances, soft foot checks, and dial indicators

API 610 TABLE 15 ALIGNMENT TOLERANCE LIMITS			
PUMP OPERATING SPEED RUNOUT	PARALLEL OFFSET TOLERANCE	ANGULAR GAP TOLERANCE	MAX PERMISSIBLE

Up to 1,500 RPM	< 0.08 mm (0.0031 in)	< 0.06 mm / 100 mm coupling	< 0.05 mm TIR
1,500 - 3,000 RPM (P-101A)	< 0.05 mm (0.0020 in)	< 0.04 mm / 100 mm coupling	< 0.03 mm TIR
Above 3,000 RPM	< 0.03 mm (0.0012 in)	< 0.02 mm / 100 mm coupling	< 0.02 mm TIR
Spacer Coupling (>100mm)	< 0.05 mm across spacer	< 0.04 mm across flanges	< 0.04 mm TIR

SOFT FOOT VERIFICATION & CORRECTION DISCIPLINE
Before conducting shaft alignment, individual motor foot coplanarity (Soft Foot) must be verified. - Maximum allowable soft foot deflection: Strictly < 0.05 mm (0.002 in) measured with dial indicator. - Maximum shim stack per foot: Strictly 3 shims (corrosion-resistant 304 SS precision pre-cut shims). - Pipe Strain Test: Loosening suction or discharge flange bolts must not deflect pump shaft > 0.04 mm.

PIPE STRAIN EXCEEDS PERMISSIBLE BOUNDARY
Excessive piping strain causes casing deflection that destroys mechanical seal faces within 72 hours. If shaft movement exceeds 0.04 mm during flange bolting, adjust spring pipe hangers immediately.

THERMAL GROWTH CALCULATIONS & DYNAMIC OFFSETS

Shaft growth formula, thermal expansion coefficients, and cold alignment targets

THERMAL GROWTH FORMULA & MATERIAL COEFFICIENTS

MATHEMATICAL FORMULATION: $\Delta L = L \cdot \alpha \cdot \Delta T$
Where:
- L = Vertical distance from centerline support base to shaft center (mm)
- α = Coefficient of linear thermal expansion (mm/mm/ deg C)
 * Carbon Steel / WCB: 11.7×10^{-6} / deg C
 * 12% Chrome / CA6NM: 11.0×10^{-6} / deg C
 * Austenitic SS / CF8M: 16.5×10^{-6} / deg C
- ΔT = Temperature differential between operating steady state and cold ambient (deg C)

COLD TARGET ALIGNMENT CALCULATION FOR P-101A TRAIN

REVERSE COLD OFFSET ERROR

Setting pump and motor concentric when cold results in severe thermal misalignment (1.0 mm vertical offset) when hot. This will cause catastrophic disc coupling fatigue shear within 48 hours.

ISO 10816-3 VIBRATION SEVERITY & DIAGNOSTICS

Vibration velocity zones, overall RMS thresholds, and FFT spectral analysis

ISO 10816-3 VIBRATION SEVERITY CLASSIFICATION (RIGID FOUNDATION)		
ZONE / STATUS	OVERALL VELOCITY (RMS)	OPERATIONAL MANDATE / DIRECTIVE

Zone A (Excellent) condition	0.0 - 2.3 mm/s	Newly commissioned or overhauled machinery
Zone B (Acceptable)	2.3 - 4.5 mm/s	Unrestricted long-term continuous refinery service
Zone C (Alert Limit) maintenance	4.5 - 7.0 mm/s	Operating degradation; schedule corrective
Zone D (Trip Level) auto-stop)	> 7.0 mm/s	EMERGENCY TRIP INITIATED (Interlock I-103
Axial Bearing Vib condition	> 4.5 mm/s	Inspect thrust bearing clearance and coupling

FFT SPECTRUM DIAGNOSTIC MATRIX FOR ROTATING MACHINERY
- 1X Running Speed Peak: Indicates mass unbalance on impeller or coupling. - 2X Running Speed Peak: Indicates mechanical misalignment or bent pump shaft. - Sub-Harmonic (0.42 - 0.48X): Oil whirl / oil whip instability in hydrodynamic sleeve bearings. - High-Frequency Noise (>2,000 Hz): Rolling element bearing race fatigue / cavitation shock waves. - Vane Pass Frequency (VPF = N_vanes * RPM): Impeller vane hydraulic interaction with volute cutwater.

CRITICAL SPEED RESONANCE RUN-UP
The P-101A first lateral critical speed is 1,820 RPM. During motor acceleration, pass through the critical speed resonance zone in less than 1.5 seconds to prevent excessive blade deflection.

ULTRASONIC THICKNESS TESTING & REMAINING USEFUL LIFE

UTM grid scanning, nominal vs retiring thickness, and corrosion rates

CORROSION MONITORING & REMAINING USEFUL LIFE (RUL) EQUATION

MATHEMATICAL FORMULATION: $RUL\ (Years) = (t_{actual} - t_{minimum}) / Corrosion_Rate$
Where:
- t_{actual} = Current ultrasonic thickness measurement (mm)
- $t_{minimum}$ = Code calculated minimum required retiring thickness (mm)
- $Corrosion_Rate$ = Measured wall thinning rate (mm / year)
CORROSION CLASSIFICATION:
- $RUL > 10$ Years: Class 1 Healthy (Inspect triennially)
- $RUL\ 5 - 10$ Yrs: Class 2 Monitored (Inspect annually)
- $RUL < 3$ Years: Class 3 Critical (Mandatory turnaround replacement spool fabrication)

UTM INSPECTION SUMMARY FOR P-101A CASING & PIPING

- Pump Casing Volute: Nominal 22.0 mm | Minimum 14.5 mm | Last UTM 19.8 mm | Rate 0.12 mm/yr | RUL: 44.1 Yrs
- Suction Piping (10-in): Nominal 12.7 mm | Minimum 6.5 mm | Last UTM 10.8 mm | Rate 0.18 mm/yr | RUL: 23.8 Yrs
- Discharge Piping (8-in): Nominal 15.2 mm | Minimum 8.2 mm | Last UTM 12.4 mm | Rate 0.22 mm/yr | RUL: 19.1 Yrs
- Minimum thickness calculations conform to ASME B31.3 Chapter II hoop stress equations.

LOCALIZED PITTING INSPECTION VIA PHASED ARRAY

Standard point UTM can miss localized naphthenic acid pitting. High-risk elbow extrados must be scanned using Phased Array Ultrasonic Testing (PAUT) across a 50x50 mm calibrated grid.

MASTER INSTRUMENT TAG DICTIONARY & DCS SIGNAL MAP

Comprehensive reference table of tags, units, calibrated spans, and trip thresholds

COMPREHENSIVE INSTRUMENT TAG DIRECTORY (REFINERY COMPLEX 01-08)

DCS CONTROLLER BLOCK ADDRESSING
All signals are wired to Triconex SIS Rack 01-SIS-A and Yokogawa Centum VP Node 01-DCS-01 via redundant optical fiber. Modbus TCP register mapping and OPC UA node addresses are published in the CogniShift sovereign instrument table.

FIELD INSTRUMENT RE-CALIBRATION NOTICE
All instrument spans must be verified using secondary deadweight testers and Hart communicators every turnaround. Zero shift exceeding 0.5% span requires immediate sensor rebuild.

EMERGENCY DIRECTORY & INCIDENT COMMAND TREE

Statutory escalation directory, disaster management hierarchy, and crisis contacts

REFINERY CRISIS MANAGEMENT & INCIDENT COMMAND HIERARCHY
TIER-1 / TIER-2 / TIER-3 DISASTER ESCALATION THRESHOLDS
- Tier 1: Localized unit incident (Minor leak or single pump trip) - Contained by shift crew. - Tier 2: Major plant incident (Uncontrolled hydrocarbon release / fire) - Mobilizes full MRPL emergency brigade. - Tier 3: Off-site disaster threat (Atmospheric H2S release exceeding fence line) - Involves District Administration.
MANDATORY STATUTORY NOTIFICATION WINDOW
Under OISD-STD-118 Section 12, any Level 2 or Level 3 incident must be officially notified to the Ministry of Petroleum and Natural Gas (MOP&NG) and OISD within exactly 24 hours of occurrence.